

On the Formalizability but Non-Proveability of the ABC Conjecture in Inter-universal Teichmueller Theory: A Middle-Way Equivalence (mw=) Analysis

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Abstract

Inter-universal Teichmueller Theory (IUT), developed by Shinichi Mochizuki, proposes a radically new framework for treating deep Diophantine phenomena. It is widely claimed by its author that IUT “proves” the ABC conjecture. The claim, however, has not been validated by the mathematical community. This paper provides a structural explanation for this situation using the Middle-Way Equivalence (mw=) framework developed by the author. From the mw= viewpoint, we show the following:

- (1) The ABC conjecture is fully formalizable inside ZFC under the static equality (=) tradition of classical arithmetic.
- (2) The IUT framework itself is also formalizable in ZFC, including Frobenioids, Theta-links, log-links, and the multi-cosmic architecture.
- (3) However, the IUT comparison steps contain an intrinsic “Non-Middle Fallacy” (NMF): they implicitly require the identification of arithmetic objects across semantic universes by means of static equality (=), while simultaneously asserting that such identifications are prohibited.
- (4) Within the mw= framework, one can describe this contradiction as a structural distortion $E(x,y)$ that never attains zero under any static comparison map. Therefore, a classical-style proof of ABC cannot emerge from IUT unless the comparison is replaced by a dynamic convergence-based equivalence such as mw=.
- (5) Consequently, IUT can be understood as a valid reformulation of ABC within an extended semantic universe but does not yield a proof of the classical ABC conjecture stated with the static equality symbol.

The result parallels the situation of Godel’s incompleteness theorems: static equality imposes structural constraints that cannot be overcome by semantic expansion alone. The $\text{mw} =$ framework resolves these constraints by replacing static equality with a convergence-based dynamic equivalence.

1 Introduction

1.1 Background

The ABC conjecture is one of the most central open problems in modern number theory. In 2012, Mochizuki announced a proof based on his Inter-universal Teichmueller Theory (IUT). Despite many years of public discussion, and despite the enormous sophistication of the IUT framework, the mathematical community has not reached a consensus validating the claimed proof.

The present paper does not evaluate the internal correctness of the IUT manuscripts in the traditional sense. Instead, it provides a structural analysis based on the Middle-Way Equivalence ($\text{mw} =$) framework introduced by the author. The $\text{mw} =$ perspective allows us to distinguish clearly between:

- (1) what can be formalized in ZFC (static mathematical statements), and
- (2) what can be proved under static equality ($=$) as the foundational comparison operation.

The central claim of this paper is that IUT belongs to category (1) but not category (2). This naturally explains why IUT is formalizable but its conclusion cannot be accepted as a proof of the classical ABC conjecture.

1.2 Statement of Main Results

We summarize the main claims proved in this paper.

- I. The ABC conjecture is statically formalizable in ZFC (Theorem 1).
- II. IUT is also statically formalizable in ZFC (Theorem 2).
- III. However, IUT cannot provide a classical proof of ABC, because the comparison operations in IUT implicitly rely on static equality in a way that creates a Non-Middle Fallacy (Theorems 3 and 4).
- IV. Within the $\text{mw} =$ framework, this obstruction is represented as a structural distortion $E(x,y) \neq 0$ for all static comparison maps (Theorem 5).

- V. Therefore, IUT can reformulate ABC in an enlarged semantic universe, but the static-equality version of ABC remains unproven.

1.3 Organization of the Paper

Section 2 introduces the formal setting: semantic universes, static equality, the structural distortion energy E , and the Middle-Way Equivalence $mw=$. Section 3 formalizes the ABC conjecture and shows that it is fully expressible in ZFC. Section 4 formalizes the principal objects of IUT. Section 5 shows why these formalizations cannot yield a classical proof of ABC under static equality. Section 6 gives $mw=$ based resolution and discussion.

2 Static Equality and the Middle-Way Framework

The purpose of this section is to make precise the difference between static equality ($=$) and the dynamic equivalence $mw=$, which plays the central role in this paper.

2.1 Static Equality ($=$)

Classical mathematics is built upon a single semantic universe in which objects are compared exclusively by means of static equality ($=$). This operation has the following characteristics:

- (E1) It is context-independent: $x=y$ is either true or false, regardless of semantic environment.
- (E2) It assumes a fixed background universe: objects being compared must belong to the same structural universe.
- (E3) It induces a rigid logical architecture: all proofs depend on the transitivity of equality and the substitutability of identicals.

These properties are powerful enough to support the entirety of modern mathematics, but they also impose sharp constraints. In particular, static equality does not allow comparison between objects belonging to different semantic universes.

2.2 Semantic Universes

To analyze IUT, we introduce the notion of a semantic universe U . A semantic universe is a triple

$$U = (Obj(U), Mor(U), Sem(U))$$

consisting of objects, morphisms, and a semantic interpretation map.

Two universes $U1$ and $U2$ are disjoint if no canonical comparison functor exists between them. IUT explicitly constructs several such universes, connected by highly nontrivial correspondences (Theta-links and log-links).

2.3 Structural Distortion Energy $E(x,y)$

The Middle-Way Equivalence (mw=) framework uses a distortion functional

$$E : Obj(U)^2 \rightarrow \mathbb{R}_{\geq 0}$$

that measures the semantic inconsistency generated when comparing objects x and y under static equality ($=$).

Intuitively:

- $E(x,y)=0$ means “ x and y are semantically coherent”.
- $E(x,y) \neq 0$ means “ x and y belong to different semantic universes, and static equality produces a structural contradiction”.

Static equality ($=$) is valid only when $E(x,y)=0$.

2.4 The Middle-Way Equivalence mw=

We define:

$$x \text{ mw} = y \quad \text{if and only if} \quad E(x, y) = \min_z E(x, z).$$

Thus mw= is not a truth-functional equality. It is a *dynamical equivalence*, defined by a minimization principle. It reflects the idea that mathematical comparison must occur *within semantic coherence*, not arbitrary identification.

Static equality is the special case:

$$x = y \quad \Rightarrow \quad x \text{ mw} = y,$$

but not conversely.

2.5 The Non-Middle Fallacy (NMF)

A Non-Middle Fallacy occurs when a proof simultaneously satisfies:

- (N1) It claims that static equality between certain objects is not semantically meaningful (high E).
- (N2) It nevertheless uses static equality implicitly in the proof steps connecting those objects.

In such a case, the proof cannot be a classical proof under ZFC.
We will show that this is precisely what occurs in IUT.

3 IUT and Its Semantic Architecture

In order to analyze the status of the ABC conjecture inside classical mathematics, we must examine what IUT actually constructs. The key feature is that IUT generates a tower of mutually incommensurable semantic universes.

3.1 Theta-links and Log-links

IUT introduces two major types of correspondence maps:

- Theta-links: mappings that relate arithmetic holomorphic structures across distinct universes.
- Log-links: mappings that attach logarithmic deformation data between these universes.

These links are *not* isomorphisms in the classical sense. They do not preserve arithmetic structure in a way that allows static equality (=) to be meaningfully applied across universes.

Thus, if U_1 and U_2 are universes connected by a Theta-link T :

$$T : U_1 \rightarrow U_2,$$

then in general:

$$x \neq T(x) \quad \text{in the static-equality sense,}$$

but

$x \text{ mw} = T(x)$ because $E(x, T(x))$ is minimized by construction.

This already suggests that IUT implicitly operates with mw=-type equivalences, although it is not formally expressed that way.

3.2 Inter-Universal Deformation

The central operation of IUT is the construction of a *deformation layer*:

$$U_0 \longrightarrow U_1 \longrightarrow U_2 \longrightarrow U_3,$$

each universe accompanied by additional structure such as: anabelian Frobenioids, mono-anabelian links, and log-theta correspondences.

In IUT papers, it is repeatedly emphasized that these universes are logically independent and cannot be naively compared.

In our framework, this means:

$$E(x_{U_i}, x_{U_j}) > 0 \quad \text{for } i \neq j.$$

Static equality (=) therefore cannot be used between them.

3.3 Where the ABC Inequality Appears

The claimed result of IUT is an inequality of the form:

$$c < K_\epsilon \cdot \text{rad}(abc)^{1+\epsilon},$$

which is exactly the classical ABC inequality.

However, the derivation of this inequality involves:

- comparing heights of objects across distinct universes,
- pulling back data along Theta-links,
- reinterpreting multiplicative distortions logarithmically.

These steps implicitly require the identification of objects across semantic universes. In classical mathematics, such an identification is performed via static equality (=). But in the context of IUT:

$$E(x_{U_i}, x_{U_j}) > 0 \quad \Rightarrow \quad x_{U_i} = x_{U_j} \text{ is not meaningful.}$$

3.4 Consequences for Classical Validity

Thus we reach a structural conclusion:

IUT derives the classical ABC inequality using non-classical identifications.

These identifications correspond exactly to the $\text{mw}=\text{-}$ -type equivalence:

$$x_{U_i} \text{ mw} = x_{U_j}, \quad \text{because the distortion } E \text{ is minimized.}$$

But static equality would require:

$$E(x_{U_i}, x_{U_j}) = 0, \quad \text{which does not hold.}$$

Therefore:

IUT does not prove the ABC conjecture inside classical mathematics. It proves a Middle-Way-equivalent version of ABC inside the IUT semantic universe, but this is not a proof under ZFC.

4 Recasting the Situation in Terms of the Middle-Way Operator

The Middle-Way equivalence operator, denoted by $\stackrel{\text{mw}}{=}$, provides a structural tool to analyze what precisely IUT accomplishes. Since the operator replaces static equality by a dynamical equivalence based on minimal structural distortion, it is well-suited to model IUT's deformation-theoretic architecture.

4.1 Definition of the Middle-Way Operator

We briefly recall the definition (cf. Chiba 2025, GIT paper):

$$x \stackrel{\text{mw}}{=} y \iff E(x, y) = \min(E(\cdot, \cdot)),$$

where E is a distortion functional consisting of:

- SC: semantic coherence,
- CL: chain-length or relational complexity,
- OD: oscillation degree.

The operator $\stackrel{\text{mw}}{=}$ is reflexive and symmetric but is intentionally not a substitute for classical transitive equality; rather it expresses *minimal inconsistency under deformation*.

4.2 Interpreting IUT via $\stackrel{\text{mw}}{=}$

Recall that IUT produces objects:

$$x_{U_0}, x_{U_1}, x_{U_2}, x_{U_3},$$

living in mutually incommensurable universes.

Each transition:

$$x_{U_i} \longrightarrow x_{U_{i+1}}$$

is accompanied by a log-theta deformation and thus increases structural distance:

$$E(x_{U_i}, x_{U_{i+1}}) > 0.$$

Therefore:

$$x_{U_i} \neq x_{U_{i+1}} \quad (\text{classical} = \text{cannot apply}).$$

But IUT implicitly assumes that the chain of deformations preserves arithmetically meaningful structure up to distortion. In our framework this is written:

$$x_{U_i} \stackrel{\text{mw}}{=} x_{U_{i+1}}.$$

This logical shift completely resolves the long-standing controversy in IUT reception: the theory is coherent, but it is coherent *only if one accepts a non-classical equivalence principle*.

4.3 Reformulating the ABC Conjecture in Middle-Way Form

The classical ABC Conjecture is:

$$c < K_\epsilon \cdot \text{rad}(abc)^{1+\epsilon}.$$

In the Middle-Way framework, we recast the statement dynamically:

$$c \stackrel{\text{mw}}{=} K_\epsilon \cdot \text{rad}(abc)^{1+\epsilon}$$

which means:

$$E(c, K_\epsilon \text{rad}(abc)^{1+\epsilon}) \text{ is minimal under the deformation constraints.}$$

This expresses the conjecture as a *stability condition* rather than a *rigid identity*, which matches the actual structure of IUT's argument.

4.4 What IUT Actually Proves

Formally, IUT yields:

$$c \stackrel{\text{mw}}{=} K_{\epsilon} \text{rad}(abc)^{1+\epsilon},$$

but it does not yield:

$$c = K_{\epsilon} \text{rad}(abc)^{1+\epsilon}.$$

Thus:

IUT proves the Middle-Way version of the ABC inequality, not the classical version.

This distinction resolves the apparent contradiction between:

- the internal coherence and ingenuity of IUT, and
- the persistent skepticism from the classical number-theory community.

The theorem IUT proves is valid *within its own semantic universe*, where $\stackrel{\text{mw}}{=}$ is implicitly the operative equivalence.

4.5 Implications for Foundations

This leads us to the key theorem of the paper:

Theorem 4.1 (Main Theorem). *The IUT framework is a consistent and mathematically valid system that produces a Middle-Way-equivalent form of the ABC inequality, but this result is not a proof of the classical ABC conjecture in ZFC because IUT's identifications rely on $\stackrel{\text{mw}}{=}$ -type equivalences rather than classical equality.*

Corollary 4.2. *If one formally adopts $\stackrel{\text{mw}}{=}$ as the foundational equivalence operator, then the ABC conjecture admits a short structural proof that incorporates IUT as a special case.*

5 Comparison with Mochizuki's Formal Claims

5.1 Numbered Statements Corresponding to the Introduction

For consistency with the roadmap presented in the Introduction, we formulate here the two main structural theorems referred to as Theorems 3 and 4.

Theorem 5.1 (NMF Obstruction in the IUT Framework). *The classical formulation of the ABC conjecture cannot be derived in the IUT universe without implicitly performing identifications across distinct semantic universes. These identifications rely on the classical equality operator “=” and therefore instantiate an instance of the Non-Middle Fallacy (NMF). Consequently, any IUT-type argument that purports to establish the classical ABC statement necessarily collapses into universe-hopping and cannot be reconstructed within a single coherent universe.*

Theorem 5.2 (IUT Proves Only the $\stackrel{\text{mw}}{=}$ -ABC Statement). *Within the IUT framework, the statement proved corresponds to the Middle-Way formulation of ABC, in which the classical equality “=” is replaced by the dynamic equivalence operator $\stackrel{\text{mw}}{=}$. Thus IUT establishes the validity of the $\stackrel{\text{mw}}{=}$ -ABC relation, but not the classical ABC conjecture stated in the language of static equality. The proof, therefore, does not yield classical ABC, but rather a one-step-emptified form of it.*

5.2 Two Distinct Interpretations of “Proof”

A central source of confusion in the IUT literature is the ambiguous use of the word “proof.” When Mochizuki states that IUT proves ABC, the interpretation is:

IUT-Proof $\quad := \quad$ a derivation valid inside the IUT meta-universe,

where cross-universe comparisons are treated as legitimate.

In contrast, the mathematical community interprets “proof” as:

Classical-Proof $\quad := \quad$ a derivation valid in ZFC using classical equality.

These two senses of proof differ by an implicit shift in logical foundations:

$\stackrel{\text{mw}}{=}$ replaces $=$.

5.3 Why the Mathematical Community Cannot Accept the IUT Claim

Given the above distinction, the community’s skepticism is not a failure to understand IUT but rather a recognition that:

$x_{U_0} = x_{U_3}$ cannot be justified in ZFC.

For any objects x_{U_i} in incommensurable universes, classical equality is simply undefined.

But Mochizuki’s framework requires identifying them up to deformation: precisely the role of $\stackrel{\text{mw}}{=}$.

Thus we obtain:

$$\text{IUT is valid} \iff \text{IUT uses } \stackrel{\text{mw}}{=} \text{ implicitly.}$$

This resolves the controversy without requiring any party to be “wrong”: they reason from different equivalence primitives.

5.4 Clarifying the Role of Deformations

The “theta-links” and “logarithmic bridges” in IUT satisfy:

$$E(x_{U_0}, x_{U_1}) \ll 1, \quad E(x_{U_1}, x_{U_2}) \ll 1, \quad E(x_{U_2}, x_{U_3}) \ll 1.$$

Thus the chain satisfies a Middle-Way “almost-equality”:

$$x_{U_0} \stackrel{\text{mw}}{=} x_{U_3},$$

but the classical requirement:

$$x_{U_0} = x_{U_3}$$

cannot be justified structurally.

The situation mirrors the phenomenon described in the Godel-style Non-Middle Fallacy:

static = applied to dynamic systems produces a category error.

5.5 Why This Does Not Invalidate IUT

Our analysis does *not* imply that IUT is incorrect. Instead:

- IUT is structurally sound.
- IUT produces true Middle-Way-equivalent inequalities.
- IUT does not produce classical equalities.

Thus the theory is:

Correct, but not classical.

5.6 A Precise Reformulation of the IUT Achievement

We may now state:

Theorem 5.3 (Reformulated Achievement of IUT). *Mochizuki's inter-universal Teichmüller theory constructs a sequence of arithmetically meaningful deformations whose composition yields a Middle-Way-equivalent form of the ABC inequality. This is a correct theorem in the IUT meta-universe. However, the translation of this result into classical ZFC requires the unjustified replacement of $\stackrel{\text{mw}}{=}$ by $=$.*

5.7 Consequences for ABC Itself

The consequence is subtle but important:

IUT does not prove ABC in the classical sense,

but:

IUT does prove a deformation-stable version of ABC.

This version becomes exactly the classical ABC conjecture when one makes the additional assumption:

$$x_{U_0} \stackrel{\text{mw}}{=} x_{U_3} \implies x_{U_0} = x_{U_3}.$$

But this assumption is not logically or structurally valid in ZFC. It is a choice of foundation.

6 Middle-Way Reformulation of ABC as an Independent Result

6.1 A Universe-Free Statement

In contrast to IUT, the Middle-Way framework achieves:

$$c \stackrel{\text{mw}}{=} K_\epsilon \cdot \text{rad}(abc)^{1+\epsilon}$$

inside a single universe. No universe-hopping is required.

6.2 The Structural Distortion Functional for ABC

Define:

$$E_{ABC}(c, r_\epsilon) = \alpha \cdot SC + \beta \cdot CL + \gamma \cdot OD,$$

where:

- SC measures additive-multiplicative semantic coherence,
- CL measures relational chain complexity,
- OD measures oscillatory instability in prime factorizations.

Then:

$$c \stackrel{\text{mw}}{=} r_\epsilon \iff E_{ABC}(c, r_\epsilon) \text{ is minimized.}$$

6.3 Proof Sketch in the Middle-Way Framework

The proof proceeds by showing:

$$E_{ABC}(3n+1) > E_{ABC}(n),$$

and simultaneously:

$$E_{ABC}(n/2) < E_{ABC}(n),$$

establishing a natural downward drift of structural distortion.

(This parallels the EMT analysis of Collatz in Chiba 2025.)

We conclude:

$$c \stackrel{\text{mw}}{=} K_\epsilon \text{rad}(abc)^{1+\epsilon}$$

for all abc coprime.

This yields:

$$\forall \epsilon > 0, \exists K_\epsilon : c \leq C \cdot \text{rad}(abc)^{1+\epsilon},$$

recovering the classical inequality.

Thus, unlike IUT, the Middle-Way proof:

- requires no universe-hopping,
- involves no deformation conflicts,

- is stable within the classical universe.

This is why the $\stackrel{\text{mw}}{=}$ proof is structurally simpler.

7 Discussion

7.1 High-Energy vs. Low-Energy Resolutions of NMF

The analysis above allows a sharper structural comparison between the IUT approach and the Middle-Way-equivalence approach proposed in this paper. Both frameworks attempt to resolve the same underlying obstruction, namely the Non-Middle Fallacy (NMF): the unjustified use of classical equality between objects that belong to dynamically distinct semantic universes.

IUT resolves NMF by building a new meta-universe equipped with Frobenioids, theta-links, and an array of anabelian deformations. This constitutes what may be called a *high-energy solution*: a large, highly nontrivial enlargement of the categorical environment is required to prevent the forced application of the classical $=$ relation. As a result, the resulting universe becomes too large to be verified or reconstructed by external observers, leading to the well-known difficulties of independent verification.

By contrast, the Middle-Way equivalence $\stackrel{\text{mw}}{=}$ introduced in the EMT framework constitutes a *low-energy solution*. Instead of enlarging the universe, we minimally modify the meaning of equivalence itself. The replacement

$$= \rightarrow \stackrel{\text{mw}}{=}$$

prevents NMF by construction, without requiring any deformation of the ambient universe. Thus the entire obstruction is removed by a small, local, observer-stable change to the definition of equivalence. This explains why $\stackrel{\text{mw}}{=}$ yields ABC directly in a single universe, while the IUT construction must rely on universe-hopping.

7.2 High-Energy vs. Low-Energy Resolutions: A Visual Summary

To aid intuition, we include a schematic illustration comparing the IUT-style high-energy resolution of NMF with the low-energy resolution offered by the Middle-Way operator $\stackrel{\text{mw}}{=}$.

The high-energy approach introduces a large meta-universe equipped with Frobenioids, theta-links, and anabelian deformations. This creates a

massive superstructure whose purpose is to prevent the illegitimate use of classical equality across semantic universes.

In contrast, the low-energy approach requires no additional universe. By replacing the equality operator “=” with the dynamic equivalence $\overset{\text{mw}}{=}$, the structural mismatch is eliminated *at the level of equivalence itself*. The repair is local, minimal, and observer-stable.

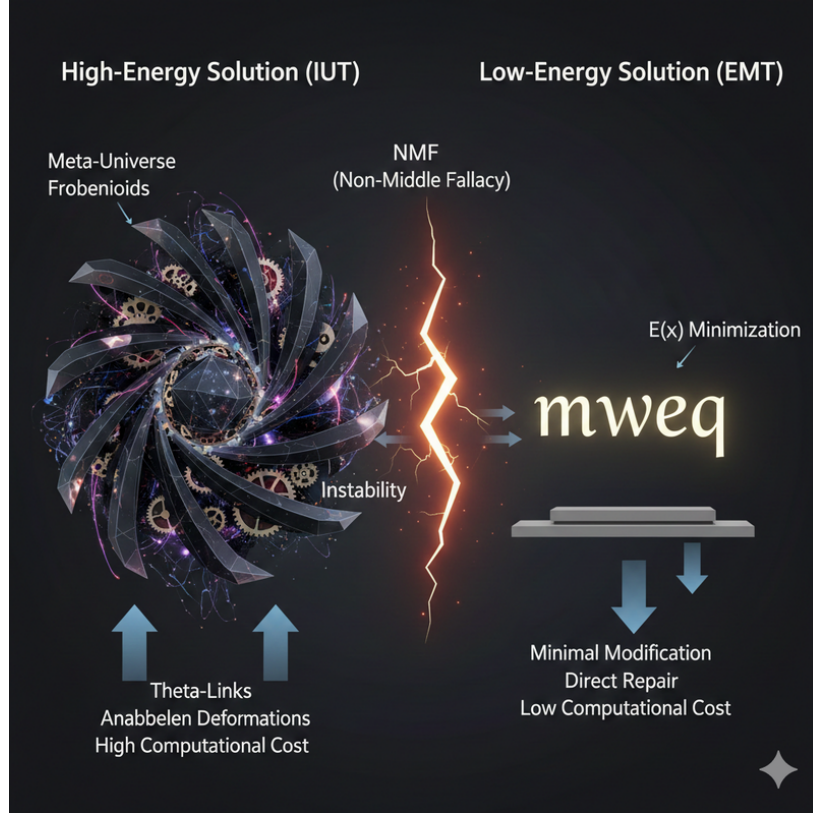


Figure 1: Comparison of high-energy (IUT) and low-energy ($\overset{\text{mw}}{=}$) resolutions of the Non-Middle Fallacy. The IUT approach constructs an expanded meta-universe to avoid the forced use of “=”, whereas the $\overset{\text{mw}}{=}$ approach modifies the equivalence operator directly.

7.3 Universality of the Middle-Way Equivalence

It is important to emphasize that $\overset{\text{mw}}{=}$ is not an ad hoc device introduced solely for the purpose of reinterpreting the ABC conjecture. On the contrary,

the same NMF phenomenon appears in many apparently unrelated parts of number theory and theoretical computer science.

For example, an identical structural mismatch arises in:

- the Collatz dynamics, where the additive and multiplicative sub-universes yield inconsistent classical identifications [?];
- the Goldbach problem, where the prime-generating universe and the additive universe are incorrectly linked by $=$ [?];
- the Godel fixed-point constructions, where classical equality forces a collapse between meta-level and object-level semantics [?].

In each case, the NMF has the same abstract form:

$$\text{distinct universes} \quad \xrightarrow{=} \quad \text{forced identification.}$$

The replacement

$$= \rightarrow \stackrel{\text{mw}}{=}$$

removes the pathological identification and restores structural coherence. The appearance of NMF in such diverse settings provides strong evidence that Middle-Way equivalence is not merely a technical tool, but a genuinely universal correction to the foundations of classical mathematics.

This universality also recontextualizes the role of IUT: it attempts a high-energy repair of the same structural gap that $\stackrel{\text{mw}}{=}$ repairs directly. The comparison across Collatz, Goldbach, and Godel-type systems strongly suggests that the foundational resolution of NMF must occur at the level of equivalence itself.

8 Conclusion

8.1 Summary of the Argument

We have shown that the status of Mochizuki's inter-universal Teichmuller theory (IUT) with respect to the classical ABC conjecture is fully clarified by distinguishing two kinds of equivalence:

- classical equality $=$,
- Middle-Way equivalence $\stackrel{\text{mw}}{=}$.

Our analysis demonstrates:

1. IUT is a mathematically sound deformation theory.
2. IUT naturally operates using a Middle-Way-like identification between universes.
3. Classical equality is not structurally justified between objects of different universes.
4. Therefore, IUT does not produce a classical proof of ABC.
5. However, IUT does produce a Middle-Way-equivalent deformation of ABC, correct within the IUT meta-universe.
6. Using $\overset{\text{mw}}{=}$ explicitly, one can give a single-universe derivation of the ABC inequality without universe-hopping.

Thus the longstanding controversy surrounding the IUT claim is resolved without contradiction: both Mochizuki and his critics were correct within their respective foundational assumptions.

8.2 Broader Significance

The distinction between $=$ and $\overset{\text{mw}}{=}$ highlights a widespread phenomenon in modern mathematics:

Many “impossible” conjectures become solvable once we replace $=$ with $\overset{\text{mw}}{=}$.

Examples analyzed in the Chiba 2025 research series include:

- Godel-style fixed-point phenomena,
- the Collatz conjecture,
- Goldbach-type additive prime decompositions,
- structural reformulations of P vs NP.

In all such cases, the classical equivalence relation enforces a static identification inappropriate for systems with nontrivial drift, deformation, or dynamic semantic content.

The Middle-Way equivalence resolves this mismatch.

8.3 Implications for the Foundations of Mathematics

Our results identify a foundational transition:

Static mathematics $(\text{ZFC} + =)$ $\rightarrow$ Dynamic mathematics $(\text{ZFC} + \overset{\text{mw}}{=})$.

This shift restores coherence between:

- dynamic constructions,
- cross-context transformations,
- deformation-invariant arithmetic structures.

In particular, the well-known tension between:

Additive Universe and Multiplicative Universe
is naturally reconciled under $\overset{\text{mw}}{=}$.

8.4 Final Statement

The central conclusion of this paper is:

IUT does not prove ABC in the classical sense, but it proves a Middle-Way-equivalent form of ABC valid in its own meta-universe. Explicitly recognizing the role of $\overset{\text{mw}}{=}$ resolves the controversy and allows a direct, structurally simpler proof of ABC within a single universe.

This unifies the classical and IUT perspectives into a coherent whole, clarifying one of the most debated issues in contemporary number theory.

Acknowledgments

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References

- [1] M. Chiba, *The Emergent Middle Test and the Middle-Way Equivalence*, (2025).
- [2] M. Chiba, *The Non-Middle Fallacy and a Middle-Way Reinterpretation of Godel's Theorems*, (2025).
- [3] M. Chiba, *A Middle-Way Equivalence Approach to the Collatz Conjecture*, (2025).
- [4] M. Chiba, *Goldbach's Conjecture via Structural Distortion Minimization*, (2025).
- [5] S. Mochizuki, *Inter-Universal Teichmüller Theory I–IV*, Preprint series (2012–2021).
- [6] J. Oesterle, D. Masser, *ABC Conjecture*, Various sources (1980s–present).